ITA0502 COMPUTER VISION

SLOT C

Railway Track Monitoring System

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Comprehensive Scenario-Based Solution Document

Prepared from the supplied assignment scenario

A. Assignment Problem / Challenge

The assignment presents a realistic computer-vision engineering challenge in which a railway safety organization must select and justify a real-time vision approach for monitoring railway tracks from cameras mounted on inspection vehicles. The source scenario explicitly requires students to apply course concepts, analyze requirements and constraints, compare alternatives, use appropriate engineering/computing tools, interpret results, and make a justified engineering decision. 毅filecite學turn0file0學L33-L38係

Scenario Context

The proposed system is OpenCV-based and must automatically detect cracks, broken components, surface defects, and abnormal objects on railway tracks. The system is safety-critical and must operate on an edge device with moderate computational resources. It must remain reliable despite vibration, motion blur, poor illumination, shadows, rust, and complex backgrounds. 毅filecite學turn0file0學L5-L12係

Engineering Challenge

The central engineering problem is not simply to choose the algorithm with the highest nominal accuracy. The selected solution must simultaneously satisfy minimum accuracy, latency, frame-rate, false-positive, computational, and robustness requirements. Therefore, the decision is a constrained multi-objective engineering selection problem.

The five candidate approaches supplied in the scenario are: Edge + Hough Transform, SIFT-based feature matching, Viola–Jones, YOLO, and a Deep Learning model. Their stated accuracy, latency, and computation levels provide the quantitative basis for the first-stage screening. 毅filecite學turn0file0學L22-L31係

Assignment Deliverables Interpreted

- Formulate the problem and define measurable objectives.
- Extract and analyze all stated operational requirements.
- Compare all five candidate approaches quantitatively and qualitatively.
- Show explicit calculations and explain the pass/fail logic.
- Consider alternative hybrid approaches where a single technique is insufficient.
- Address environmental robustness, safety, computational limits, and deployment constraints.
- Select a final deployable architecture and provide recommendations for validation.

B. Problem Statement

Problem / Case / Design Challenge:

Design a real-time OpenCV-based railway track monitoring system that processes camera frames from an inspection vehicle and automatically identifies cracks, broken components, surface defects, and abnormal objects. The deployed system must achieve detection accuracy of at least 92%, latency of no more than 100 ms/frame, processing speed of at least 15 FPS, false-positive rate of no more than 5%, and moderate computational demand suitable for edge deployment. It must also be robust to vibration, motion blur, poor illumination, shadows, rust, and complex backgrounds.
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Objective Definition

Primary objective: select the safest practical computer-vision approach or combination of approaches that satisfies the hard operational constraints while maintaining adequate robustness and moderate edge-device resource usage.

Secondary objectives: minimize unnecessary false alarms, preserve real-time operation, provide interpretable detection evidence for maintenance personnel, and establish a validation process that can demonstrate performance under representative railway conditions.

Decision Principle

Because railway inspection is safety-critical, accuracy and robustness are prioritized. A method that fails a hard constraint should not be selected merely because it is fast or computationally inexpensive. The source explicitly states that accuracy and robustness must be prioritized. 毅filecite孳turn0file0孳L13-L21像

Scope

- Included: algorithm selection, quantitative screening, hybrid architecture, edge-processing strategy, validation logic, constraints, assumptions, safety considerations, and deployment recommendations.
- Excluded: procurement-specific hardware selection, railway regulatory certification, trackside installation design, and creation of a new training dataset because those details are not provided in the scenario.

C. Requirements and Constraints

The source specifies explicit operational thresholds and additional qualitative constraints. These are treated as acceptance criteria rather than soft preferences. 毅filecite學turn0file0學L13-L21僚

Requirement / Constraint	Required condition	Interpretation
Detection accuracy	≥ 92%	Hard safety/performance threshold; below this the system fails.
Latency	≤ 100 ms/frame	Hard real-time processing threshold; above this the system fails.
Processing speed	≥ 15 FPS	Minimum real-time inspection rate.
False-positive rate	≤ 5%	Hard acceptability threshold; above this the system is unacceptable.
Computation	Moderate / edge-suitable	High computation is not preferred.
Environment	Vibration, blur, low light, shadows, rust, complex backgrounds	Robustness must be demonstrated, not assumed.
Safety	Safety-critical inspection	Accuracy and robustness receive highest decision weight.
Data/implementation	OpenCV-based	Solution should be implementable in an OpenCV-centered pipeline.

Candidate Performance Data

Approach	Accuracy	Latency (ms/frame)	Approx. speed from latency*	Computation	Initial status
Edge + Hough Transform	88%	35	28.57 FPS	Low	Reject
SIFT-based feature matching	90%	60	16.67 FPS	Moderate	Reject
Viola–Jones	84%	45	22.22 FPS	Low	Reject
YOLO	94%	85	11.76 FPS	Moderate	Does not meet 15 FPS if latency is per frame
Deep Learning model	97%	130	7.69 FPS	High	Reject

*The FPS values above are derived from the supplied per-frame latency using FPS = 1000 / latency(ms). This derived calculation is used only as an engineering consistency check; the source separately states a required processing speed of at least 15 FPS. The source does not provide measured FPS values for the candidates. 毅filecite學turn0file0學L8-L10僚

Hard vs. Soft Constraints

Hard constraints are accuracy ≥92%, latency ≤100 ms/frame, FPS ≥15, and false-positive rate ≤5%. Computational demand and environmental robustness are critical deployment considerations, with safety taking precedence. Importantly, the scenario does not provide false-positive rates for the five candidates; therefore, no candidate can honestly be declared compliant with the ≤5% false-positive requirement from the supplied numerical data alone.

D. Student Work

1. Problem Understanding and Formulation

1.1 What is the problem?

The problem is to convert camera imagery captured from a moving inspection vehicle into timely and reliable maintenance/safety information. Railway imagery differs from a static object-detection benchmark because the camera platform itself moves and vibrates, causing blur, viewpoint changes, illumination changes, and background variation. Track components may also be rusty, partially occluded, dirty, or visually similar to surrounding materials.

1.2 Expected Outcomes

- A deployable real-time detection pipeline.
- Quantitative evidence that the chosen approach meets or is capable of meeting the stated thresholds.
- An explicit rejection rationale for unsuitable approaches.
- A robustness and validation plan for railway environmental conditions.
- A safe decision strategy that avoids treating uncertain detections as confirmed failures.

1.3 Available Information / Data

Data item	Available in scenario	Use in analysis
Target defects/objects	Yes	Defines detection classes and task scope.
Accuracy threshold	Yes	Primary screening criterion.
Latency threshold	Yes	Real-time feasibility.
FPS threshold	Yes	Operational throughput.
False-positive threshold	Yes	Safety/acceptability.
Computation level	Qualitative	Edge deployment feasibility.
Environmental conditions	Yes	Robustness assessment.
Candidate accuracy/latency/computation	Yes	Alternative comparison.
Candidate false-positive rates	No	Must be measured during validation.
Hardware specification	No	Exact FPS/latency must be benchmarked on target edge hardware.

1.4 Assumptions

- Accuracy figures are comparable across candidates and represent the same evaluation task.
- Latency is interpreted as end-to-end per-frame processing time for the relevant candidate.
- FPS derived from latency is an idealized upper bound when frames are processed sequentially and no additional pipeline overhead exists.
- The final implementation can use a combination of classical preprocessing/verification and a learned detector, provided the combined latency remains within limits.
- False-positive performance must be experimentally established because it is absent from the supplied candidate data.

1.5 Constraints

The constraints are simultaneously performance, computational, environmental, and safety constraints. The system cannot trade away a hard requirement without becoming unsuitable for the stated application.

2. Application of Course Knowledge

2.1 Relevant Principles

- Image preprocessing: denoising, contrast normalization, illumination correction, and region-of-interest extraction.
- Edge detection and Hough Transform: useful for geometric structures such as rails and track boundaries.
- Feature extraction/matching: useful for correspondence against known reference patterns, but sensitive to viewpoint and texture changes.
- Object detection: bounding-box based localization of abnormal objects/components.
- Deep learning: strong representation learning for complex visual patterns, at the cost of computational demand.
- Performance engineering: latency, throughput, memory, and compute must be measured on the deployment target.

2.2 Mathematical Models and Calculations

The basic throughput relationship is:

$$\text{FPS} = 1 \,/\, \text{T} \;\; (\text{seconds/frame}) = 1000 \,/\, \text{T\_ms}$$

For each candidate, this converts the supplied latency into an approximate maximum sequential throughput.

Approach	Calculation	Derived FPS	FPS ≥ 15?
Edge + Hough	1000 / 35	28.57	Yes
SIFT	1000 / 60	16.67	Yes
Viola–Jones	1000 / 45	22.22	Yes
YOLO	1000 / 85	11.76	No
Deep Learning	1000 / 130	7.69	No

2.3 Margin to Each Hard Threshold

Accuracy margin is calculated as candidate accuracy − 92%. Latency margin is 100 − candidate latency. FPS margin is derived FPS − 15.

Approach	Accuracy margin	Latency margin (ms)	FPS margin
Edge + Hough	−4%	+65	+13.57
SIFT	−2%	+40	+1.67
Viola–Jones	−8%	+55	+7.22
YOLO	+2%	+15	−3.24
Deep Learning	+5%	−30	−7.31

This margin analysis shows the fundamental trade-off. The fastest candidates fail accuracy; the highest-accuracy candidates fail real-time latency/throughput; YOLO is closest to a practical balance but, using the supplied 85 ms/frame latency literally, does not reach 15 FPS in a simple sequential implementation.

2.4 Constraint Satisfaction Logic

$$\text{Feasible} = (\text{Accuracy} \geq 92\%) \text{ AND } (\text{Latency} \leq 100 \text{ ms}) \text{ AND } (\text{FPS} \geq 15) \text{ AND } (\text{FPR} \leq 5\%)$$

Since the scenario supplies no FPR values, the complete feasibility condition cannot be proven for any candidate from the given data. The correct engineering conclusion is therefore 'no candidate is fully validated by the supplied evidence'; however, candidates can still be ranked by the available evidence.

3. Solution / Design / Methodology

3.1 Alternative 1 — Edge + Hough Transform

Edge detection followed by the Hough Transform is attractive for extracting strong geometric structures such as rails and track boundaries. Its supplied latency of 35 ms/frame is excellent for edge deployment, and its low computation is favorable. However, its 88% accuracy is 4 percentage points below the mandatory 92% threshold. It should therefore not be the sole detector.

3.2 Alternative 2 — SIFT-Based Feature Matching

SIFT can provide robust local features and matching under moderate scale and rotation changes. Its 90% accuracy is still below the 92% requirement, while 60 ms/frame gives only a 40 ms latency margin and a derived 16.67 FPS—just 1.67 FPS above the minimum. Environmental changes such as rust, blur, and altered texture can also reduce stable correspondences. It is better treated as a verification cue than as the primary safety detector.

3.3 Alternative 3 — Viola–Jones

Viola–Jones offers low computation and 45 ms/frame latency, making it fast enough in principle. Nevertheless, its 84% accuracy is 8 percentage points below the required threshold. Its classical cascade structure is also less naturally suited to diverse railway defects and complex backgrounds. It should be rejected as the principal detector.

3.4 Alternative 4 — YOLO

YOLO has the highest accuracy among the candidates that remain within the stated 100 ms latency limit: 94% accuracy and 85 ms/frame. It therefore clears the accuracy and latency thresholds. The difficulty is throughput: 85 ms/frame corresponds to approximately 11.76 FPS, below the 15 FPS requirement. This makes YOLO the strongest candidate in accuracy/latency balance, but it requires optimization, pipelining, frame skipping with temporal tracking, or a faster edge inference configuration before final acceptance.

3.5 Alternative 5 — Deep Learning Model

The deep learning model has the best stated accuracy at 97%, but its 130 ms/frame latency violates the 100 ms hard limit and its high computation is contrary to moderate-resource edge deployment. Its derived throughput is only 7.69 FPS. It should therefore be rejected for direct deployment in the supplied form, even though its accuracy is attractive.

3.6 Comparative Decision Matrix

Approach	Accuracy	Latency	FPS*	Compute	Hard-constraint view	Decision
Edge + Hough	88%	35 ms	28.57	Low	Accuracy fail	Reject as primary
SIFT	90%	60 ms	16.67	Moderate	Accuracy fail	Reject as primary
Viola–Jones	84%	45 ms	22.22	Low	Accuracy fail	Reject
YOLO	94%	85 ms	11.76	Moderate	FPS fail*	Best base; optimize
Deep Learning	97%	130 ms	7.69	High	Latency + FPS fail	Reject direct

*FPS is calculated from latency because the source gives latency but not measured FPS for each candidate. Therefore this is a derived engineering estimate, not an experimentally reported metric.

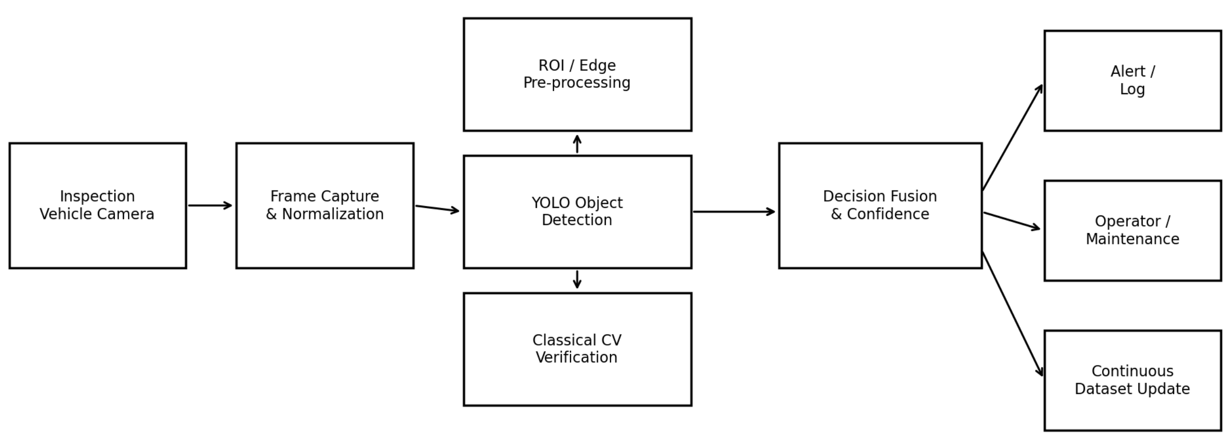
3.7 Proposed Combination — Optimized YOLO + Classical CV

The most appropriate deployment strategy is a hybrid architecture centered on YOLO, supplemented by lightweight classical computer-vision stages. YOLO supplies the learned semantic detection capability needed for complex defects and abnormal objects. Edge/ROI processing can constrain the search region and reduce unnecessary computation. Hough-based geometry can validate rail alignment/track regions, while SIFT-style matching can be used selectively for high-value verification rather than on every frame.

3.8 Why a Hybrid Strategy?

- YOLO is the only supplied candidate that clears the 92% accuracy threshold while also remaining below 100 ms/frame.
- Edge/ROI processing can reduce the image area presented to the detector.
- Geometric checks can reject detections outside the physical rail/track region.
- Selective classical verification can reduce false alarms without forcing expensive processing on every frame.
- Temporal consistency across adjacent frames can stabilize detections under vibration and motion blur.
- The design retains a clear path to moderate-resource edge deployment if profiling demonstrates the required 15 FPS.

Proposed Edge Railway Track Monitoring Architecture



3.9 Step-by-Step Processing Pipeline

Step 1 — Frame acquisition: Capture frames from the inspection-vehicle camera with synchronized timestamps. Record camera parameters and frame dimensions.

Step 2 — Stabilization / motion compensation: Estimate global camera motion where feasible and use lightweight stabilization to reduce the effect of vehicle vibration.

Step 3 — ROI extraction: Identify the track/rail region and crop irrelevant background. This reduces inference workload and limits false detections outside the inspection area.

Step 4 — Illumination normalization: Apply contrast normalization or controlled brightness correction. Avoid aggressive enhancement that amplifies rust texture or sensor noise.

Step 5 — YOLO inference: Run the optimized YOLO detector on the normalized ROI. Quantization, smaller input resolution, or hardware acceleration should be evaluated while preserving accuracy.

Step 6 — Classical verification: Apply geometry/edge checks to validate that a candidate lies within a physically plausible rail or track region. Use feature matching selectively when a reference comparison is useful.

Step 7 — Temporal fusion: Track detections over successive frames. Require consistent evidence across a short temporal window before raising a maintenance alert, except for clearly hazardous abnormal objects.

Step 8 — Confidence and alert decision: Combine detector confidence, geometric consistency, and temporal persistence. Route high-confidence events to logging/alerting.

Step 9 — Data logging: Store frame ID, timestamp, location metadata if available, class, confidence, and evidence image for later review.

Step 10 — Human review / maintenance workflow: Safety-critical alerts should support inspection personnel rather than silently replacing engineering judgment. Uncertain cases are flagged for review.

3.10 False-Positive Control Strategy

- Use a physically defined track ROI so background objects are not treated as track defects.
- Use class-specific confidence thresholds tuned on validation data.
- Use temporal persistence to suppress one-frame noise.
- Use geometry checks for rail-associated classes.
- Maintain a hard evaluation target of FPR ≤ 5%; do not claim compliance until measured on a representative test set.

3.11 Latency Budget

Pipeline stage	Target budget (ms)	Engineering rationale
Capture + preprocessing	10–15	Fast image preparation.
ROI / geometry	5–10	Lightweight classical operations.
YOLO inference	55–70	Main compute allocation.
Post-processing + tracking	5–10	NMS, temporal fusion, logging.
Safety margin	5–15	Absorb device/OS variation.
Total target	≤100	Must be benchmarked on target hardware.

The table is a proposed engineering budget, not a measurement from the source scenario. It demonstrates how the 100 ms ceiling can be treated as a design budget. A target below 100 ms is preferable because real deployments include operating-system, memory-transfer, camera, and logging overhead.

4. Use of Modern Tools

4.1 Recommended Toolchain

Tool	Purpose	Evidence/output
Python	Pipeline implementation and experiment orchestration	Scripts, benchmark logs
OpenCV	Capture, preprocessing, edges, Hough, ROI, visualization	Processed frames and timing logs
YOLO implementation	Primary object detection	Bounding boxes, confidence, class metrics
NumPy	Numerical/image-array operations	Preprocessing and evaluation calculations
Git	Version control and experiment traceability	Commit history, tags
Edge-device profiler	CPU/GPU/NPU utilization and latency	Resource/latency report

4.2 Experimental Workflow

1. Freeze the candidate dataset split before tuning.
2. Measure baseline accuracy, latency, FPS, and false-positive rate for each candidate.
3. Profile preprocessing, inference, post-processing, and total pipeline time separately.
4. Optimize the YOLO candidate while recording every accuracy/latency change.
5. Test robustness separately under blur, low illumination, shadows, rust, vibration, and complex backgrounds.
6. Evaluate false positives by class and environmental condition.
7. Run a final locked test set only after all thresholds are fixed.
8. Document the hardware, software version, image resolution, batch size, and inference configuration.

4.3 Reproducibility Requirements

- Record exact model weights/version and configuration.
- Record image resolution and preprocessing operations.
- Record target edge hardware and accelerator settings.
- Use fixed evaluation datasets and deterministic seeds where practical.
- Report both average and high-percentile latency, not only the fastest frame.
- Keep failed experiments because they explain engineering trade-offs.

5. Results and Validation

5.1 Quantitative Screening Result

Based on the supplied values, the candidate ranking is not a simple accuracy ranking. Deep Learning ranks first in accuracy but fails latency and computation. YOLO ranks second in accuracy and remains within the latency ceiling, but its derived throughput falls below 15 FPS. The three classical methods are fast enough but fail the 92% accuracy requirement.

Candidate	Accuracy test	Latency test	FPS test	Compute test	Overall conclusion
Edge + Hough	FAIL (88%)	PASS	PASS*	PASS	Not acceptable alone
SIFT	FAIL (90%)	PASS	PASS*	Moderate	Not acceptable alone
Viola–Jones	FAIL (84%)	PASS	PASS*	PASS	Not acceptable
YOLO	PASS (94%)	PASS (85 ms)	FAIL* (11.76)	PASS/conditional	Best candidate for optimization
Deep Learning	PASS (97%)	FAIL (130 ms)	FAIL* (7.69)	FAIL	Reject for supplied deployment form

*Derived from supplied latency. Actual FPS must be measured on the target edge device. No candidate can be declared fully compliant because the source does not supply candidate false-positive rates.

5.2 Validation Test Matrix

Condition	Test design	Primary metric	Acceptance target
Normal daylight	Representative track images/video	Accuracy, FPR, FPS	Accuracy ≥92%, FPR ≤5%, FPS ≥15
Poor illumination	Low-light sequences	Accuracy, recall, FPR	No unacceptable degradation
Shadows	Strong cast-shadow sequences	FPR, accuracy	FPR ≤5%
Rust / aged components	Rust-heavy track samples	Accuracy, FPR	Accuracy ≥92%, FPR ≤5%
Motion blur	Controlled/real vehicle blur	Recall, localization	Operationally acceptable detection
Vibration	Vehicle-motion sequences	Latency, stability	FPS ≥15; stable temporal detections
Complex backgrounds	Objects near tracks	FPR	FPR ≤5%
Long-duration run	Continuous inspection route	Latency distribution, failures	No sustained real-time failure

5.3 Statistical Reporting

A robust evaluation should report confusion matrices, per-class precision and recall, overall accuracy as defined by the chosen evaluation protocol, false-positive rate, missed-defect rate, and confidence intervals where dataset size permits. For latency, report mean, median, 95th percentile, and worst-case or maximum observed latency. A system averaging 80 ms/frame but frequently exceeding 100 ms is risky for a hard latency requirement.

5.4 Acceptance Gate

$$\text{Accept only if: Accuracy} \geq 92\%,\ \text{Latency} \leq 100\ \text{ms/frame},\ \text{FPS} \geq 15,\ \text{and FPR} \leq 5\%$$

If any hard threshold fails, the system returns to optimization or redesign rather than being accepted by averaging the failed metric away.

6. Analysis and Engineering Decision

6.1 Advantages and Limitations of Alternatives

Approach	Advantages	Limitations	Engineering role
Edge + Hough	Very fast; low compute; geometric interpretability	88% accuracy; limited semantic understanding	ROI/rail geometry verification
SIFT	Moderate compute; useful local correspondence	90% accuracy; sensitive to blur/texture changes; narrow FPS margin	Selective reference verification
Viola–Jones	Fast; low compute	84% accuracy; weak fit for diverse defects	Not recommended for safety-critical primary detection
YOLO	94% accuracy; 85 ms latency; moderate compute	Derived 11.76 FPS; FPR unknown	Primary detector after optimization
Deep Learning	97% accuracy	130 ms; high compute; 7.69 FPS derived	Offline benchmark / future optimized model

6.2 Trade-Off Analysis

The principal trade-off is accuracy versus real-time efficiency. Classical methods satisfy speed more comfortably but miss the required accuracy. The Deep Learning model provides the strongest accuracy but exceeds the latency and compute envelope. YOLO occupies the most promising middle ground, with only a 2-point accuracy margin and a 15 ms latency margin, but it needs throughput optimization.

6.3 Final Engineering Decision

The recommended solution is an optimized YOLO-centered hybrid pipeline: YOLO for semantic detection, lightweight OpenCV preprocessing and ROI extraction for efficiency, edge/Hough geometry for physical consistency checks, and temporal fusion for stability. The final system should not be accepted until the optimized implementation is experimentally demonstrated to achieve ≥15 FPS and ≤5% false-positive rate on the target edge device and representative railway data.

6.4 Approaches That Must Be Rejected

- Edge + Hough Transform — reject as the sole deployment detector because 88% accuracy is below the 92% requirement.
- SIFT-based feature matching — reject as the sole detector because 90% accuracy is below the 92% requirement.
- Viola–Jones — reject because 84% accuracy is substantially below the required threshold.
- Deep Learning model as supplied — reject because 130 ms/frame exceeds the 100 ms limit and computation is high.
- Unoptimized YOLO as supplied — do not accept yet because 85 ms/frame implies only 11.76 FPS under sequential processing; optimization and target-hardware benchmarking are required.

7. Broader Considerations

7.1 Safety

Railway inspection is safety-critical. A false negative can allow a real defect to remain undetected, while excessive false positives can overload maintenance teams and reduce trust in the system. The system should therefore be designed as a decision-support and alerting mechanism with clear confidence, evidence, audit logs, and escalation rules.

7.2 Ethics and Professional Responsibility

The engineering team must not report nominal benchmark performance as field readiness. Performance should be disclosed by operating condition, class, and hardware configuration. Known limitations should be documented, especially when the system encounters environmental conditions outside the validated dataset.

7.3 Environment

Edge processing can reduce dependence on continuous high-bandwidth transmission because frames can be processed locally and only relevant events or compact logs need to be transmitted. This may reduce communication overhead, although the actual energy impact depends on hardware and duty cycle.

7.4 Economics

A moderate-resource edge solution can reduce recurring cloud compute and communication costs. However, safety-critical deployment should prioritize lifecycle reliability and maintainability over the lowest initial cost.

7.5 Accessibility and Human Factors

Maintenance personnel should receive clear event labels, confidence levels, image evidence, timestamps, and location information where available. Alerts should be prioritized so that operators can distinguish urgent hazards from lower-confidence inspection observations.

7.6 Security and Privacy

The design should protect stored inspection imagery and logs from unauthorized modification. Model and software versions should be controlled so that an unverified update cannot silently change safety-critical behavior.

7.7 Sustainability

Model optimization, ROI processing, and event-driven storage can reduce compute and storage demand. Reusing a common OpenCV pipeline and maintaining versioned models can also improve maintainability across the system lifecycle.

8. Conclusion

8.1 Proposed Solution

A hybrid, YOLO-centered railway track monitoring architecture is the most appropriate direction. YOLO is the strongest supplied candidate because it is the only approach that simultaneously clears the stated accuracy threshold and remains below the 100 ms latency threshold. However, its 85 ms/frame latency implies 11.76 FPS if processed sequentially, so optimization is mandatory before deployment.

8.2 Major Findings

- Edge + Hough is fast and inexpensive but fails accuracy.
- SIFT is close to the accuracy threshold but still fails it and has little FPS headroom.
- Viola–Jones is computationally attractive but fails accuracy by a large margin.
- YOLO provides the best balance but needs throughput optimization.
- Deep Learning provides the highest accuracy but violates latency and moderate-compute requirements.
- False-positive compliance cannot be established from the supplied scenario because candidate FPR values are not provided.

8.3 Achievement of Requirements

Requirement	Current evidence	Status
Accuracy ≥92%	YOLO = 94%	Pass at candidate level
Latency ≤100 ms/frame	YOLO = 85 ms	Pass at candidate level
FPS ≥15	85 ms ⇒ 11.76 FPS sequentially	Fail pending optimization
FPR ≤5%	Not supplied	Unverified
Moderate computation	YOLO stated moderate	Conditionally favorable
Environmental robustness	Qualitative requirement only	Must be field-validated

8.4 Limitations

- No dataset size, class distribution, or train/test split is provided.
- No false-positive rates are supplied.
- No target edge hardware specification is supplied.
- The accuracy definition is not specified in the source.
- Derived FPS from latency may differ from actual measured FPS because pipeline concurrency and overhead are unknown.
- Environmental robustness is a requirement, not a measured result in the source.

8.5 Possible Improvements

- Use a smaller or quantized YOLO model and benchmark accuracy loss versus latency gain.
- Use hardware acceleration available on the chosen edge device.
- Reduce input ROI and resolution only after checking small-defect recall.
- Use temporal tracking to avoid full inference on every frame where safety analysis permits.
- Collect hard-negative examples from rust, shadows, ballast, sleepers, and complex backgrounds.
- Perform condition-specific threshold calibration to control false positives.

9. Student Reflection

9.1 What did you learn from this assignment beyond what was directly taught in the classroom?

This scenario demonstrates that computer-vision engineering is a constrained system-design problem rather than an isolated algorithm-selection exercise. The highest accuracy model is not automatically the best engineering solution. A useful system must balance accuracy, latency, throughput, compute, environmental robustness, and safety. The analysis also shows the importance of distinguishing measured evidence from derived estimates and from assumptions.

9.2 If you were given additional time/resources, what would you improve and why?

With additional time and resources, the first improvement would be to collect a representative railway dataset containing real vibration, motion blur, illumination, rust, shadows, occlusion, and background variation. The second would be to benchmark several optimized YOLO configurations directly on the target edge hardware. The third would be to establish a formal false-positive and false-negative evaluation by defect class. Finally, long-duration field trials would be used to assess stability, thermal behavior, memory usage, and maintenance workflow integration.

9.3 Proposed Future Work Plan

Phase	Work	Exit criterion
1. Data	Collect/label representative video	Validated dataset and class definitions
2. Baseline	Benchmark five candidate approaches	Comparable accuracy/latency/FPR report
3. Optimization	Optimize YOLO + ROI + preprocessing	≥15 FPS target with acceptable accuracy
4. Robustness	Stress-test environmental conditions	No unacceptable degradation
5. Field trial	Long-duration inspection run	Stable operation and actionable alerts
6. Acceptance	Independent validation	All hard constraints demonstrated

10. References

The following source is the primary basis for the scenario, terminology, requirements, candidate performance values, assignment structure, and evaluation expectations.

9. ITA0502 COMPUTER VISION, SLOT C — Railway Track Monitoring System, supplied assignment PDF, 6 pages.
10. OpenCV documentation and implementation resources, to be consulted during actual implementation and tool validation.
11. YOLO implementation documentation/model documentation, to be cited according to the exact framework and model version used in the experimental implementation.
12. Any railway safety standards, organizational procedures, datasets, and edge-hardware documentation used during a real deployment must be added to the final project reference list.

Appendix A — Complete Calculation Sheet

A.1 FPS Derivation

Edge + Hough Transform: FPS = 1000 / 35 = 28.57 FPS.

SIFT-based feature matching: FPS = 1000 / 60 = 16.67 FPS.

Viola–Jones: FPS = 1000 / 45 = 22.22 FPS.

YOLO: FPS = 1000 / 85 = 11.76 FPS.

Deep Learning model: FPS = 1000 / 130 = 7.69 FPS.

A.2 Accuracy Gap

Edge + Hough Transform: Accuracy margin = 88% − 92% = -4 percentage points.

SIFT-based feature matching: Accuracy margin = 90% − 92% = -2 percentage points.

Viola–Jones: Accuracy margin = 84% − 92% = -8 percentage points.

YOLO: Accuracy margin = 94% − 92% = +2 percentage points.

Deep Learning model: Accuracy margin = 97% − 92% = +5 percentage points.

A.3 Latency Margin

Edge + Hough Transform: Latency margin = 100 − 35 = +65 ms.

SIFT-based feature matching: Latency margin = 100 − 60 = +40 ms.

Viola–Jones: Latency margin = 100 − 45 = +55 ms.

YOLO: Latency margin = 100 − 85 = +15 ms.

Deep Learning model: Latency margin = 100 − 130 = -30 ms.

Appendix B — Final Decision Checklist

- □ Candidate model accuracy independently verified ≥92%.
- □ End-to-end latency independently verified ≤100 ms/frame.
- □ Sustained processing speed independently verified ≥15 FPS.
- □ False-positive rate independently verified ≤5%.
- □ Environmental robustness tested under vibration and motion blur.
- □ Low-illumination and shadow robustness tested.
- □ Rust and complex-background false positives quantified.
- □ Target edge-device compute, memory, temperature, and power profiled.
- □ Model/software versions frozen and auditable.
- □ Safety review completed before operational deployment.